

Assignment 3 - MDA 9159

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Fall 2025

- **This assignment is due Friday, Nov 28, 2025, at 11:55 pm.**
- You must write your answers and R code using Rmarkdown (template provided) and generate a single PDF file. **Submissions not generated by Rmarkdown will not be graded and receive zero marks.**
- Submissions must be done via Gradescope. You must carefully assign questions to their corresponding pages. Submissions without questions assigned to pages will **not** be graded. **Questions with no pages assigned to them will receive zero marks.**
- Always show all your work and add comments to your code explaining what you are doing.
- **Students are encouraged to create their own work and articulate their answers and explanations in their own words. It's essential to understand that scores are not based on the language used, as long as the core concepts and meanings are clearly conveyed. Therefore, using AI tools (like ChatGPT) to generate responses and directly copying them is not permitted. The focus of the assignment is to demonstrate your understanding, not that of the AI. This approach will also serve as valuable preparation for the upcoming closed-book midterm test.**

Q1. Weighted Least Squares (WLS) – 20 points

Generate heteroscedastic data: `wls_dat`

```
library(MASS)
library(car)
library(leaps)
library(glmnet)
library(caret)
library(ggplot2)

set.seed(9159)
n <- 200
x <- runif(n, 0, 10)
sigma2 <- x^2+runif(n)
y <- 1 + 4.8*x + rnorm(n, sd = sqrt(sigma2))
wls_dat <- data.frame(y, x)
```

- (a) Fit OLS $y \sim x$. Plot residuals vs fitted values and comment on heteroscedasticity.
- (b) Fit WLS with weights $w_i = 1/x^2$. Compare OLS and WLS coefficients, SEs, and CIs.
- (c) plot the regression lines from both OLS and WLS on the same scatter plot of the data and their prediction confidence intervals (in R: `interval="prediction"`), and comment on differences.
- (d) Fit WLS with misspecified weights $w_i = 1/(0.5 + 0.05x_i)^2$. Comment on the standard deviation of the estimate of the slope by comparing to OLS and WLS with correct-form of weights.

Q2. Correlated errors – 20 points

One can load the data (on BrightSpace, week 11) by the command: `p219 = read.table("P219.txt", h=T)`.

The variables are - H: Housing Starts - P: Population Size of 22- to 45-yr-olds in millions - D: Availability for Mortgage Money Index

- (a) Fit the linear model $H \sim P$. Plot residuals vs fitted values and comment on correlation structure of residuals.
- (b) Plot ACF of residuals and comment on correlation structure.
- (c) Consider adding D as a predictor. Fit the model $H \sim P + D$. Plot residuals vs fitted values and ACF of residuals. Comment on whether adding D helps to mitigate correlation in errors.
- (d) Run a Durbin-Watson test on both models and comment on the results.

Q3. Multicollinearity (mtcars) – 30 points

Use:

```
dat <- mtcars  
m_full <- lm(mpg ~ disp + hp + wt + qsec, data = dat)
```

- (a) Report correlation matrix among numeric predictors. Identify $|r| > 0.8$.
- (b) Compute **VIF**; identify moderate ($\text{VIF} > 5$) and severe ($\text{VIF} > 10$) collinearity.
- (c) Fit a reduced model to mitigate collinearity; compare adjusted R^2 and SE of `disp`.

Q4. Model Selection with Subset Selection and Regularization – 30 points

Using the built-in dataset **mtcars**, we will predict the miles per gallon (mpg) using all other variables as predictors.

- (a) Split the data into training (70%) and test (30%) sets, using
`set.seed(9159); train_index_mtcars <- sample(1:nrow(mtcars), 0.7*nrow(mtcars))`.
- (b) Perform best subset selection on the training set and identify the model with the lowest BIC. Report the selected variables and the test MSE. (try using `regsubsets` function from `leaps` package)
- (c) Fit ridge and lasso regression models on the training set using cross-validation to select optimal lambdas. Report the selected variables (for lasso) and test MSEs for both models. (try using `cv.glmnet` function from `glmnet` package)